

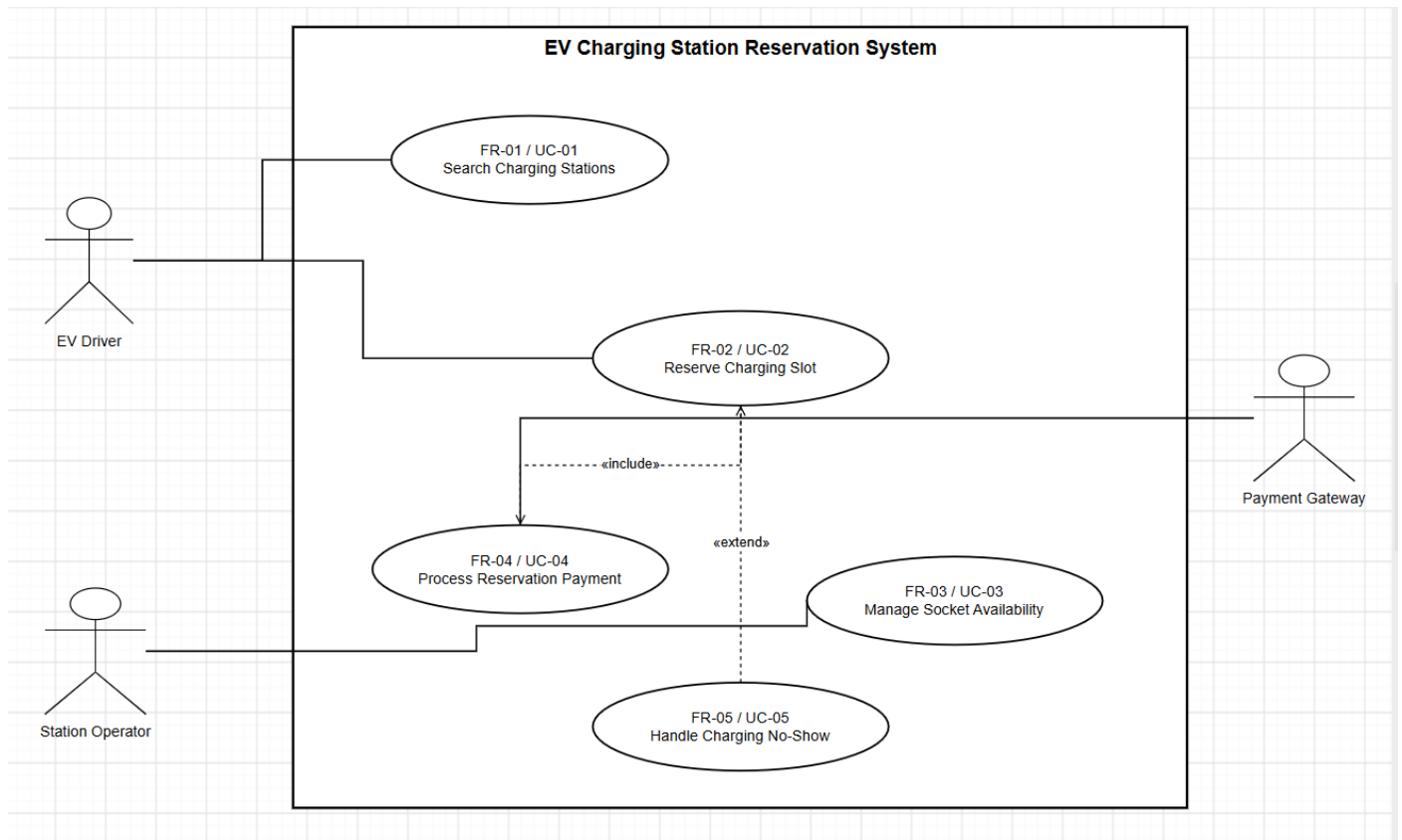
SE Lab 1

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UML Diagram



Use Case ID: UC-02

Use Case Name: Reserve Charging Slot

Primary Actor: EV Driver

Secondary Actors: Payment Gateway (external system)

Preconditions

- The EV Driver is logged in to the reservation portal.
- The driver has completed a station search and selected a station with at least one connector-compatible, available socket (see UC-01: Search Charging Stations).

Postconditions

- Success: A 30-minute charging slot is locked exclusively for the driver's vehicle, payment is authorized, and a confirmation with slot start time is issued.
- Failure: No slot is reserved, no payment is captured, and the socket remains available to other drivers.

Main Success Scenario

1. EV Driver selects "Reserve" on a connector-compatible, available charger at the chosen station.
2. System matches the driver's stored vehicle connector type (CCS, CHAdeMO, or Type 2) against the charger and confirms compatibility (FR-001).
3. System calculates the dynamic power rate for the requested time slot based on current demand and time of day (includes UC: Calculate Dynamic Power Rate — FR-003).
4. System displays the 30-minute slot window, quoted rate, and total estimated cost to the driver.
5. Driver confirms the reservation and selects a payment method.
6. System invokes payment processing and connects to the Payment Gateway (includes UC: Process Payment).
7. Payment Gateway authorizes the payment.
8. System locks the socket for the driver's vehicle for the reserved 30-minute window and starts the 10-minute plug-in timer.
9. System displays a confirmation screen with station location, socket ID, and slot start time, and emails/SMS's the confirmation to the driver.
10. Use case ends successfully.

Alternate Flow

7a. Payment Declined

- 7a1. System displays a "payment failed" message and does not lock the socket.
- 7a2. Driver is prompted to try another payment method.
- 7a3. If payment fails again after 2 attempts, the reservation is cancelled, the socket remains available, and the driver is notified.

8a. Driver Fails to Plug In Within 10 Minutes (extends UC-02 via "Apply No-Show Penalty")

- 8a1. System detects that the 10-minute plug-in timer has elapsed with no charging session started.
- 8a2. System automatically releases the locked socket, making it available to other drivers.
- 8a3. System applies a no-show penalty charge to the driver's account and sends a notification (NFR-001).

Req Id	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall match the driver's vehicle connector type (CCS, CHAdeMO, Type 2) with available station chargers and reserve a 30-minute charging window.	High	Pass: Incompatible connector plugs are hidden from the driver and only matching sockets are shown. Fail: Two vehicles are booked for the same socket concurrently.	Prevents wasted trips to incompatible stations and guarantees the driver a usable slot.
FR-002	Functional	The system shall allow an EV driver to search for nearby charging stations filtered by connector type,	High	Pass: Search results list only stations within the selected radius that have at least one available, connector-compatible socket, sorted by distance. Fail: A station	Core discovery function; drivers need to quickly find a station they can actually use before

		distance, and real-time socket availability.		with zero available compatible sockets appears in the results.	attempting a reservation.
FR-003	Functional	The system shall calculate and display a dynamic power rate (price per kWh) for a charging slot based on real-time demand and time of day before the driver confirms the reservation.	Medium	Pass: The quoted rate shown at booking matches the rate applied at the confirmation step for the same slot and timestamp. Fail: The rate charged at confirmation differs from the rate quoted during booking.	Enables demand-based pricing for the operator while giving the driver cost transparency before committing.
FR-004	Functional	The system shall allow an EV driver to cancel a confirmed reservation up to 5 minutes before the slot start time without incurring a penalty.	Medium	Pass: A cancellation submitted more than 5 minutes before slot start releases the socket immediately and no penalty is charged. Fail: A penalty is charged for a cancellation made before the 5-minute cutoff.	Gives drivers flexibility and frees up sockets promptly for other drivers, improving station utilization.
FR-005	Functional	The system shall allow a Station Operator to update a charger's status (Available, In Use, Faulty, Under Maintenance) in real time, immediately removing it from driver search and reservation results when unavailable.	High	Pass: Within 5 seconds of an operator marking a charger 'Faulty', the charger no longer appears as bookable to drivers. Fail: A driver can reserve a charger flagged 'Faulty' or 'Under Maintenance'.	Prevents drivers from being routed to non-functional chargers and lets operators manage station uptime.
NFR-001	Non- Functional	The system shall automatically release a locked socket and apply a no-show penalty if the driver fails to plug in within 10 minutes of the reserved slot start time.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load, with the socket auto-released and penalty applied exactly at the 10-minute mark. Fail: A socket remains locked beyond 10 minutes with no plug-in, or is released early.	Protects station throughput by preventing a single no-show driver from blocking a socket indefinitely.
NFR-002	Non- Functional	The system shall support at least 500 concurrent reservation requests with a 95th-percentile API response time under 2 seconds.	Medium	Pass: Load testing with 500 concurrent simulated users shows 95th-percentile response time below 2 seconds and zero dropped requests. Fail: Response time exceeds 2 seconds at the 95th percentile or requests are dropped under load.	Ensures the platform stays responsive during peak commuting hours (e.g., morning/evening rush) when many drivers search and book simultaneously.